

Project WABBIT

Wireless Autonomous Behavioral Bot with Intelligence and Traits

Overview

Project WABBIT is a portable AI personality and robotics framework designed to separate the robot's body from its mind.

The Raspberry Pi serves as the physical body, handling perception, speech synthesis, motor control, and telemetry collection.

The mobile application serves as the cognitive layer, managing memory, personality, autonomous behaviors, AI interaction, and remote operation.

This architecture allows the same WABBIT identity to be transferred between multiple robotic platforms while preserving memories, personality traits, and behavioral tendencies.

System Architecture

Raspberry Pi (Body)

The Raspberry Pi is responsible for low-level hardware interaction.

Responsibilities

- Wake Word Detection
- Speech-to-Text (Whisper)
- Text-to-Speech (Piper)
- Bluetooth Communication
- Motor Control
- Sensor Integration
- Telemetry Collection
- Autonomous Behavior Generation through ROS2

The Raspberry Pi does not directly communicate with AI providers.

Its purpose is to act as a body that perceives and acts.

Mobile Application (Brain)

The mobile application serves as the robot's cognitive layer.

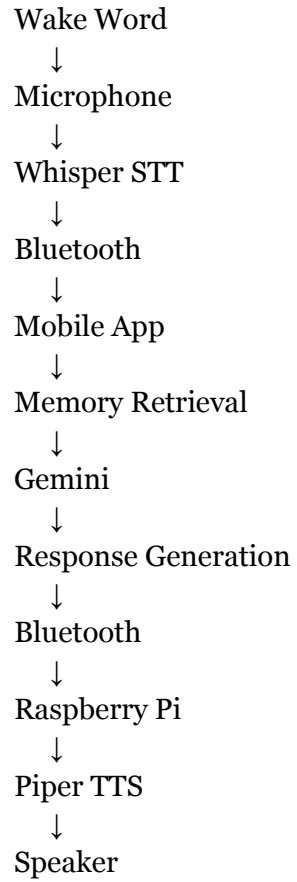
Responsibilities

- Gemini API Integration
- Long-Term Memory
- Personality System
- Telemetry Monitoring
- Remote Operation
- Conversation History
- Voice Configuration

The application acts as WABBIT's persistent mind.

Communication Pipeline

Autonomous Conversation



Personality System

WABBIT personality traits influence behavior generation rather than directly controlling motor outputs.

Curiosity Controls how frequently WABBIT generates new goals. Low Curiosity • Rarely explores • Long idle periods High Curiosity • Frequently generates goals • Constantly investigates	Exploration Controls the maximum radius used when generating new goals. Low Exploration • Remains close to its current position High Exploration • Generates distant goals • Covers larger areas
Persistence Controls how long WABBIT will continue pursuing a goal before abandoning it. Low Persistence • Gives up easily • Frequently changes goals	Energy Controls movement speed and activity levels. Low Energy • Slow movement • Reduced activity High Energy • Fast movement • Frequent actions
Sociability Controls how often WABBIT attempts to initiate interaction.	Friendliness Controls conversational tone.
High Persistence • Continues attempting difficult goals	Humor Controls conversational playfulness.

Autonomous Exploration System

WABBIT generates goals based on personality parameters.

Goal Generation

```
target = random_point(  
    radius=exploration_radius  
)
```

The Raspberry Pi transmits its set point and its current point and when or if it reaches that point.

Exploration Visualization

The application contains a procedural visualization of WABBIT's internal behavioral state.

This visualization serves as a live representation of WABBIT's intentions and progress.

Display Elements

WABBIT Icon

Represents the robot's current position.



Goal Marker

Represents the current target.



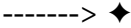
Visualization Cycle

Goal Generated



A new target is selected.

Goal Pursuit



The WABBIT icon moves toward the goal using telemetry updates.

Goal Reached



Goal completion is displayed.

New Goal

A new target is generated according to WABBIT's personality settings.

The cycle repeats continuously.

Telemetry System

The Raspberry Pi periodically reports:

```
{  
  "x": 0.25,  
  "y": -0.33,  
  "goal_x": 0.8,  
  "goal_y": 0.2,  
  "battery_voltage": 11.8,  
  "cpu_temp": 54.1,  
  "memory_usage": 42,  
  "signal_strength": -55  
}
```

The mobile application uses this data to update the exploration visualization and system status displays.

Mobile Application Screens

Screen 1: Personality & Behavior

Purpose:

Configure WABBIT's behavioral traits and observe autonomous behavior.

Features:

- Curiosity Slider
 - Exploration Slider
 - Persistence Slider
 - Energy Slider
 - Sociability Slider
 - Friendliness Slider
 - Humor Slider
 - Voice Selection
 - AI Model Selection
 - Exploration Visualization
-

Screen 2: Operator Console

Purpose:

Directly operate WABBIT without AI involvement.

Features:

Live Speech Transcript

Displays incoming speech recognized by Whisper.

Manual Text Input

Send text directly to Piper TTS.

Motor Control

Independent left and right motor controls.

Telemetry Dashboard

Displays:

- Battery Voltage
- CPU Temperature
- Memory Usage
- Wireless Signal Strength
- Connection Status

Voice System

WABBIT uses Piper for local speech synthesis.

Voice selection occurs during installation.

Example:

```
{  
  "voice": "en_GB-alba-medium"  
}
```

Future versions may support:

- Custom Trained Voices
- Personality-Based Voice Selection

Long-Term Memory

Conversation history and behavioral settings are stored locally on the mobile device.

The memory system maintains:

- Conversations
- Personality Settings
- User Preferences
- Behavioral Statistics

This allows WABBIT to retain continuity across sessions.

Technology Stack

Raspberry Pi

- Python
- Whisper
- Piper
- Bluetooth
- SQLite

Mobile Application

- Python
- Flet
- Gemini API
- SQLite

Planned Project Structure

wabbit_app/

├── main.py

├── screens/

│ ├── personality.py

│ ├── operator.py

│ └── settings.py

├── widgets/

│ ├── wabbit_map.py

│ ├── telemetry.py

│ └── motor_control.py

├── services/

│ ├── gemini_service.py

│ ├── bluetooth_service.py

│ ├── memory_service.py

│ ├── telemetry_service.py

│ └── personality_service.py

├── models/

│ ├── telemetry.py

│ ├── personality.py

│ ├── conversation.py

│ └── goal.py

├── assets/

│ ├── rabbit.png

│ └── icons/

├── config/

│ └── app_config.json

└── database/

└── wabbit.db

Long-Term Vision

Project WABBIT (Wireless Autonomous Behavioral Bot with Intelligence and Traits) is an experimental robotics and artificial intelligence platform designed to explore the concept of persistent, portable AI personalities. The system is intended to perceive, remember, learn, communicate, and autonomously explore through a robotic platform while maintaining a consistent identity across deployments and hardware configurations.

A primary objective of the project is to investigate an architecture in which the AI "mind" is separated from the physical "body." In this design, cognitive functions such as memory, personality, behavioral planning, and language processing operate independently from the robotic hardware. The robot itself is responsible for sensing, actuation, speech input and output, and interaction with the physical environment, while higher-level reasoning and behavioral systems are managed through a separate software layer.

This separation provides several advantages, including reduced computational requirements on robotic platforms, improved transparency and observability of AI decision-making, simplified migration between different robotic bodies, and greater flexibility when upgrading hardware. By decoupling intelligence from the physical platform, WABBIT can preserve its memories, personality traits, and behavioral characteristics while operating through a variety of devices and embodiments.

The long-term vision of Project WABBIT is to develop a portable artificial personality capable of maintaining continuity of memory, behavior, and interaction across multiple generations of hardware. In this framework, intelligence is treated as a transferable asset, while robotic hardware serves as a replaceable interface to the physical world. This approach enables ongoing experimentation with cognitive architectures, personality systems, and human-robot interaction while maintaining a persistent and evolving digital identity.